

No. of Functional Features Included in the Solution

Date	05 July 2026
Team ID	XXXXXX
Project Name	HDI Predictor (Human Development Index Prediction Using Machine Learning)
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Input Form	Accepts HDI-related input values from the user	Frontend (HTML Form)	Done	High
2	Input Validation	Validates user input before prediction	Flask Backend	Done	Medium
3	ML Model Prediction	Predicts Human Development Index using trained ML model	Machine Learning Model	Done	High
4	Model Loading	Loads saved hdi_model.pkl for prediction	Flask Backend	Done	Medium
5	Prediction Display	Displays predicted HDI value on webpage	HTML/Jinja Template	Done	High
6	Dataset Processing	Reads and preprocesses HDI dataset	Pandas	Done	Medium
7	Model Training	Trains regression model using dataset	Scikit-learn	Done	High
8	Responsive Web Interface	Simple user-friendly prediction interface	HTML/CSS	Done	Medium

Feature Summary:

Metric	Count / Value
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Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	User Input, ML Prediction, Result Display, Model Loading

Additional / Nice-to-Have Features	Responsive UI, Input Validation
Features Tested & Verified	All 8 features

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Inpu Form, Result Display
2	Backend / Logic	3	Flask Processing, Model Loading, Prediction Logic
3	Database / Storage	1	CSV Dataset, Saved ML Model (.pkl)
4	API / Integration	1	Flask Route Integration
5	Security / Authentication	0	Not Implemented